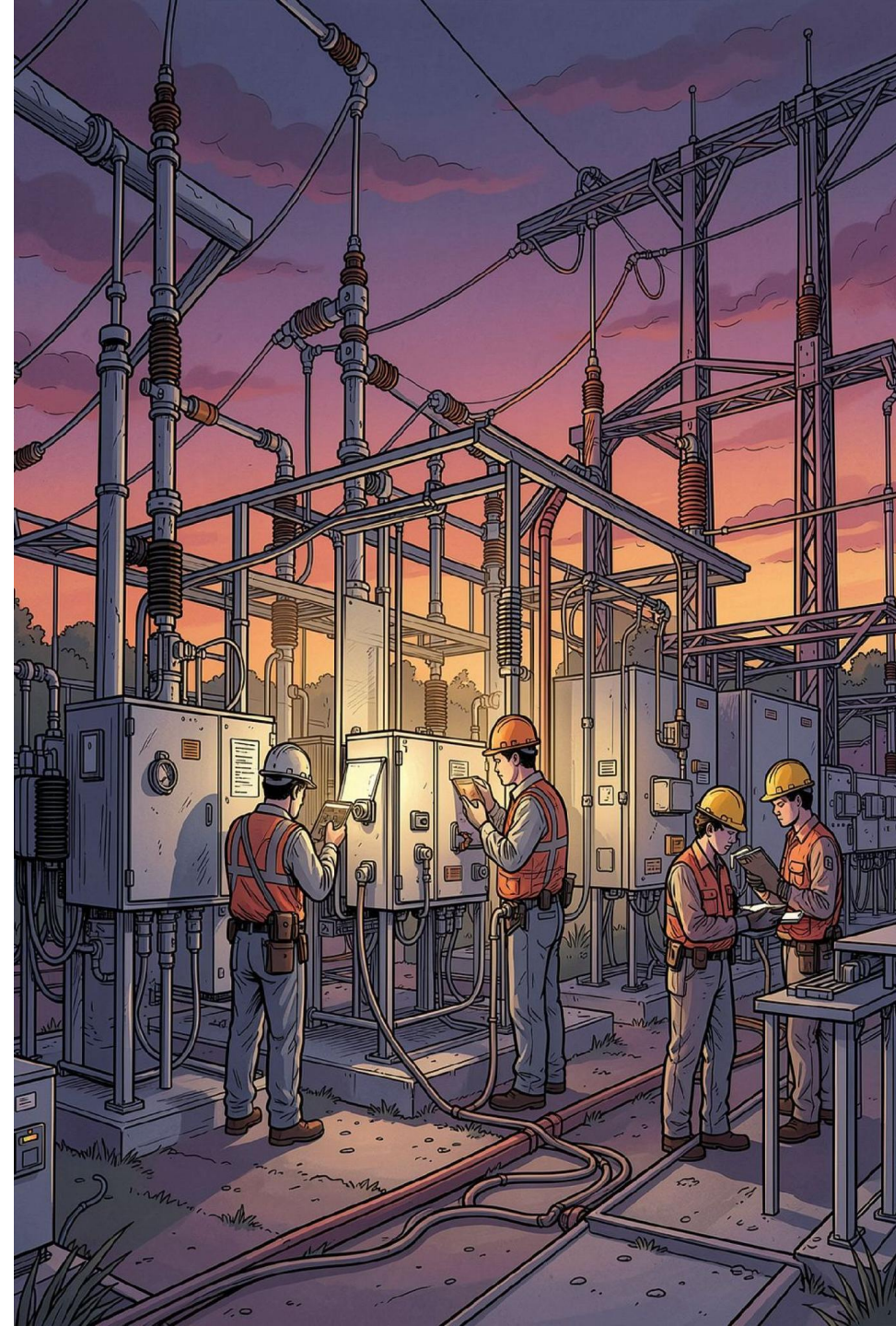
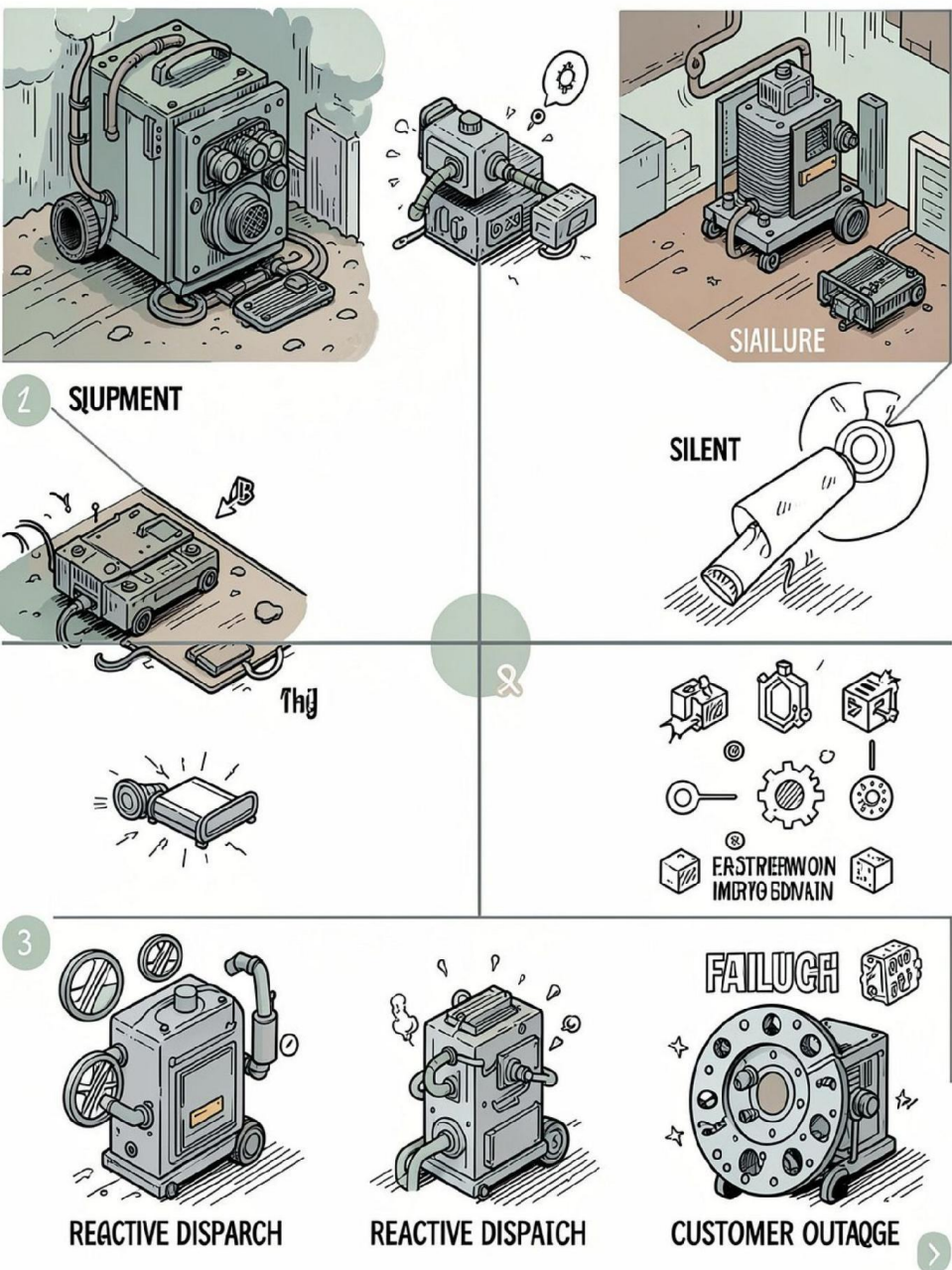


GridGuard AI — Operational Intelligence for Power Utilities

IBM Bob Hackathon 2025 | Team Photon

Turn failure predictions into immediate field actions: predict, explain, assign, and resolve —
in under 1 second.





The Problem: Power Grids Fail Without Warning

Utilities operate with incomplete signals. Equipment ages, load and weather interact, and failures appear suddenly. Operators react — dispatching crews manually, often to the wrong location or with incorrect skills — yielding longer outages and higher customer impact.

- Failure drivers: age, load, heat, weather exposure, prior incidents
- Data siloed: no unified operational decision signal
- Outcome: reactive dispatch, wrong crews, delayed repairs, customer outages

Why Existing Approaches Fall Short



Prediction alone doesn't solve operations. A probability score without recommended action, location-aware assignment, and explainability creates noise, not decisions.

Common practices

Fixed schedules, manual inspections, spreadsheets, expert intuition

What's missing

Failure probability → no action plan, no crew pre-positioning, no network impact scoring, limited explainability

The GridGuard AI Value Proposition

Predict. Assess. Prioritize. Assign. Act. GridGuard AI converts equipment, weather, and incident signals into explainable, operational field assignments in under one second — actionable intelligence, not just forecasts.



Fast

End-to-end decision in **<1s** for each asset



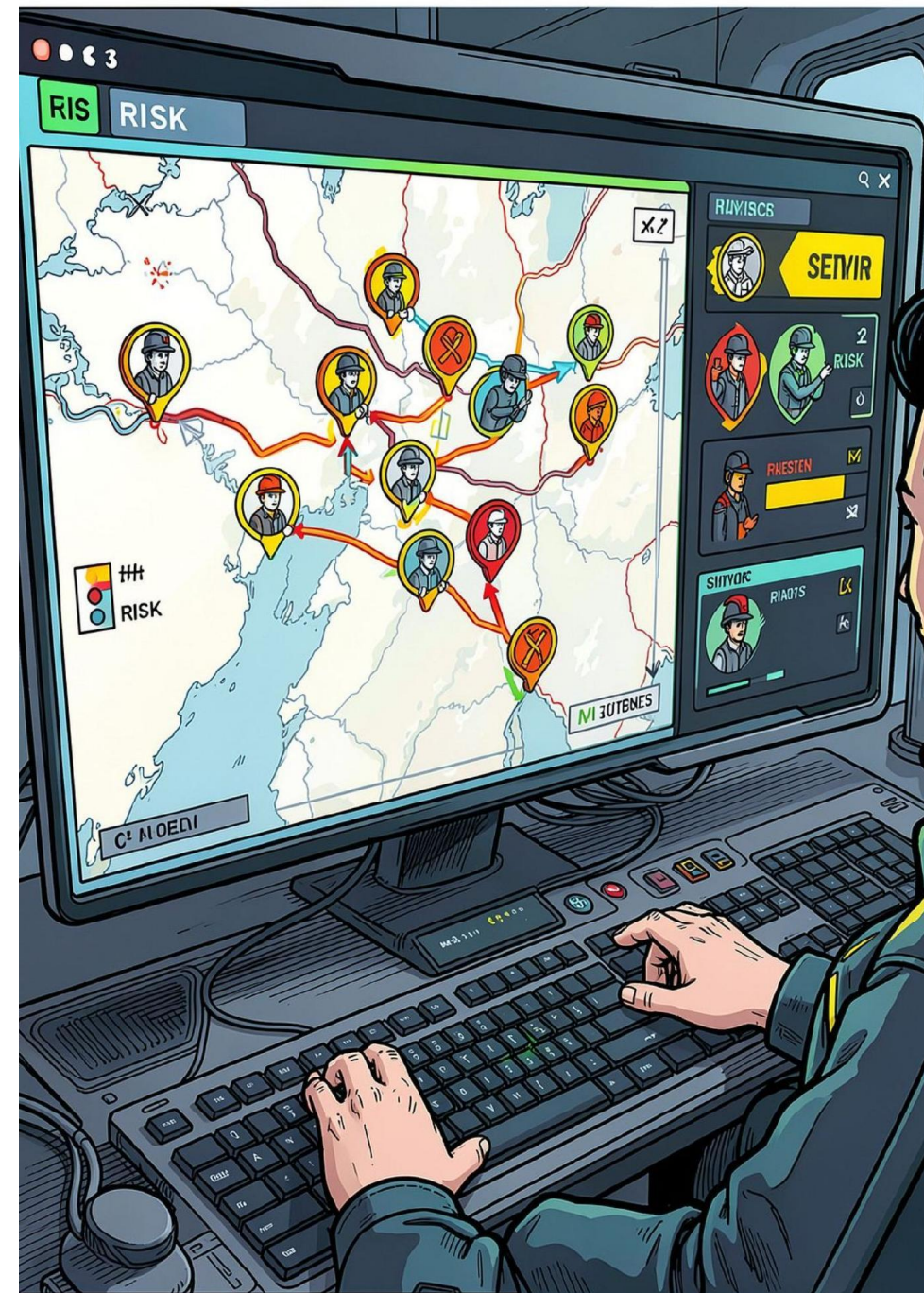
Explainable

Deterministic rules + feature importance for every recommendation



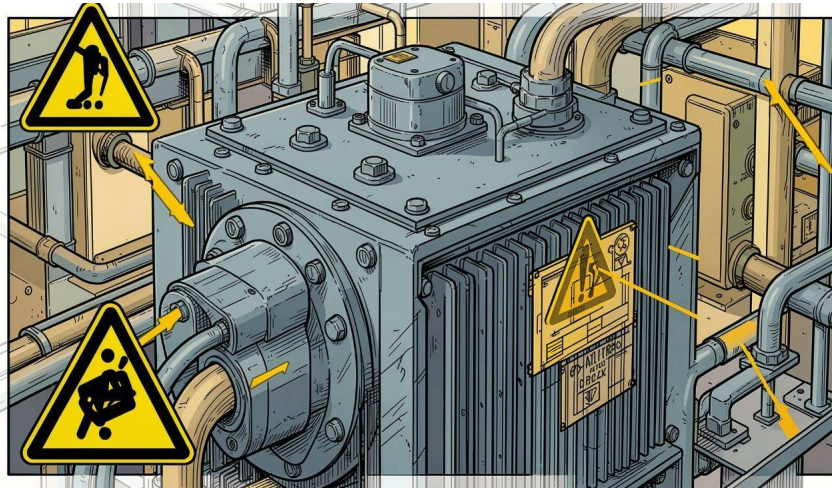
Operational

Skill-aware, distance-minimized crew assignment



Example Output: Asset T001 (Decision Card)

Clear decision card translating model and risk signals into field action.



Failure Probability

92%

Weather Risk

30 / 100

Grid Impact

83 / 100

Priority Score

77 / 100 — Risk: HIGH

Action & Crew

Preventive maintenance → Crew
C008 (8.5 km)

Why

Aging unit, high temp, high load,
11 prior incidents, critical facility

System Architecture — End-to-End Pipeline



Data ingestion (equipment, weather, incidents, crews) → feature engineering → RandomForest prediction → weather & impact scoring → priority + explainability → Haversine-based crew matching → Streamlit dashboard (maps, AI cards, simulator).

Machine Learning Design

RandomForest

Model

RandomForestClassifier · target:
failure_next_7_days ·
class_weight=balanced ·
random_state=42

21

Features

21 engineered features across
equipment, incidents, and weather

Top

Top Features

health → age → load → weather_storm
→ rainfall

Feature categories: equipment (age, health, load, voltage, customers_served), incidents (previous_incidents, severity, days_since_last_failure), weather (rainfall, wind, storm, temp stress), engineered flags (age_risk_flag, health_risk_flag, temp_stress) and network importance.

📌 Design choices prioritize reproducibility and operational stability over experimental complexity.

Risk & Decision Engine: Formulas and Thresholds

Transparent, weighted scoring that converts model output into operational priority.

1

Weather Risk (0–100)

$0.40 \times \text{rainfall_risk} + 0.30$
 $\times \text{wind_risk} + 0.20$
 $\times \text{temperature_stress} + 0.10$
 $\times \text{storm_risk}$

2

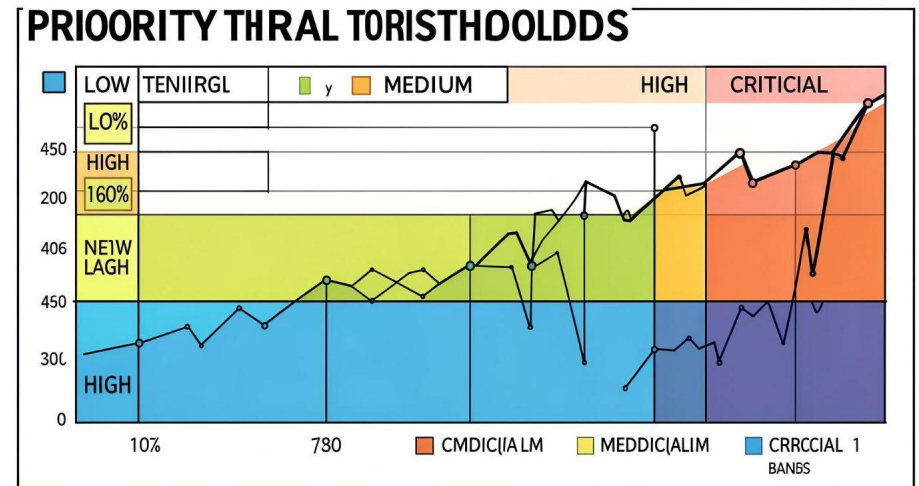
Grid Impact (0–100)

$0.40 \times \text{customer_impact} + 0.20$
 $\times \text{critical_facility} + 0.20$
 $\times \text{load_impact} + 0.20$
 $\times \text{network_importance}$

3

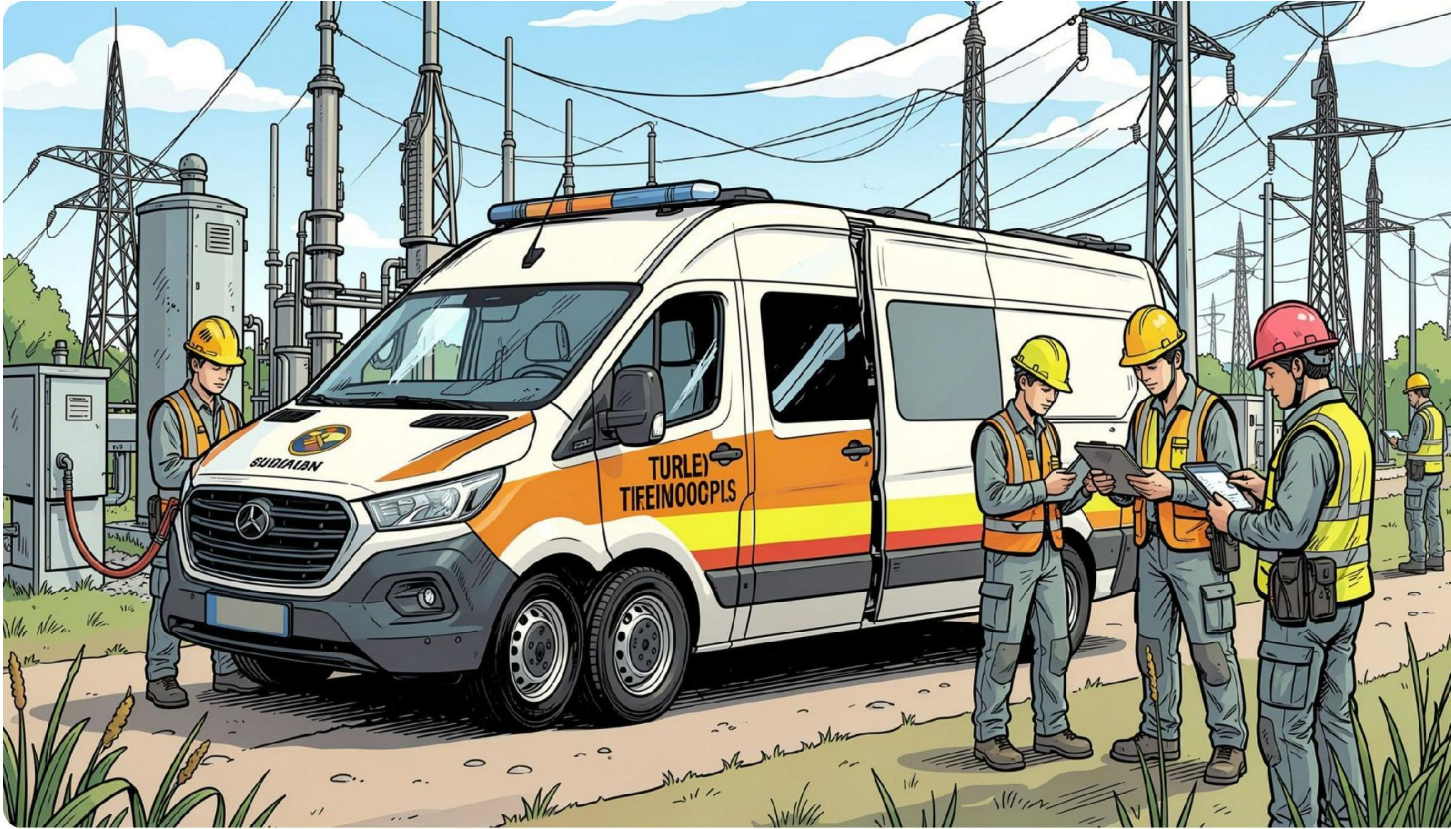
Priority Score (0–100)

$0.50 \times \text{failure_probability} + 0.20 \times \text{weather_risk} + 0.30 \times \text{grid_impact}$



Thresholds: LOW 0–30 · MEDIUM 30–60 · HIGH 60–80 · CRITICAL 80–100. Network importance is derived from voltage, customers served, and critical facility flags — computed deterministically for auditability.

Crew Pre-Positioning & Assignment Logic



Assignment triggers only for HIGH and CRITICAL assets. The algorithm filters by availability and skill, checks capacity, computes Haversine distance, and selects the nearest qualified crew. This reduces travel time and mismatch risk.

Example: T001 → C008 (8.5 km); S001 → C001 (1.5 km); L001 → C004 (19.2 km). 9/9 high-risk assets assigned in the demo run.



Skill Match

Electrical → Transformer, Switch



Mechanical

Switch, Line



Civil

Line, Transformer

Dashboard, Development Workflow & Next Steps

Dashboard 7 Pages

Overview, Risk Command Center, Live Map, Equipment AI Card, Crew Ops, AI Insights, Alert Center

Production Roadmap

Real-time SCADA ingestion, calibrated historical models, topology graph, multi-crew dispatch optimization, mobile field app, watsonx operator Q&A

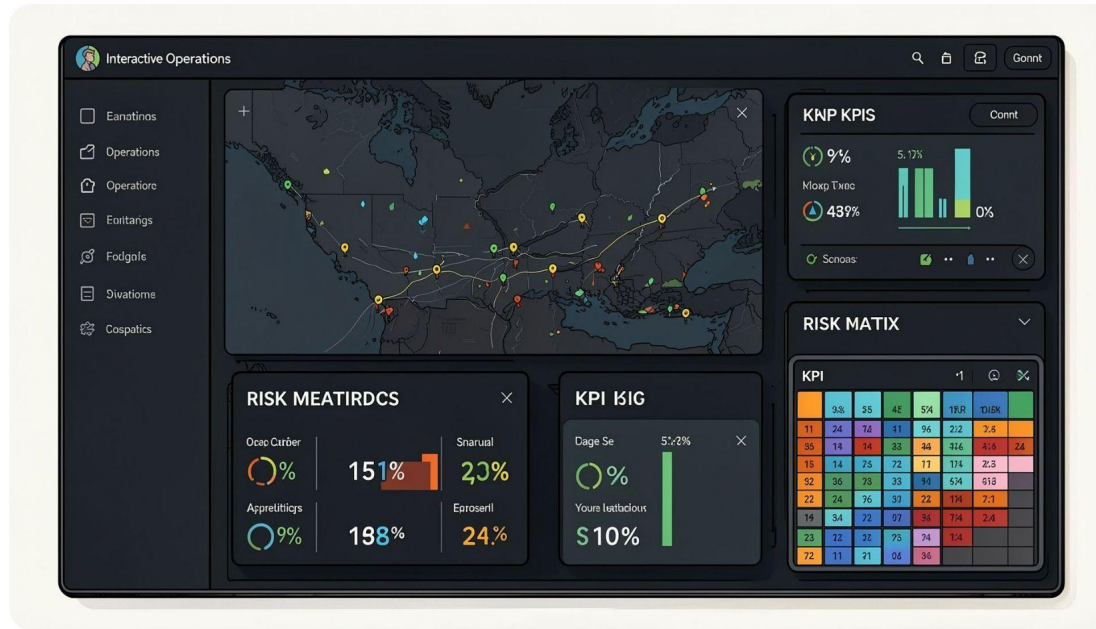
1

2

IBM Bob Role

Used as AI SDLC partner for design, code generation, testing and docs — NOT a runtime dependency

3



❏ Core story: DATA → AI → RISK → IMPACT → PRIORITY → EXPLANATION → MAINTENANCE → CREW → ACTION

Contact — Team Photon (IBM Bob Hackathon 2025). Ready to pilot: production integrations and operator acceptance testing.